

CYNTHIA OWOLABI

Senior Software Engineer | Payments & Distributed Systems

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Portfolio & engineering case studies: adebolaowolabi32.github.io

Senior software engineer with 9+ years of experience across payments, distributed applications, cloud infrastructure and applied AI. Built operational interfaces, background workflows and service integrations at Duplo, Bitso, Microsoft, Indeed and Interswitch. Experience includes reducing API scan completion from 15 to 5 hours and delivering transit software for fare collection, ticketing and fleet tracking.

TECHNICAL FOCUS

Backend & data: TypeScript/Node.js, Java, Python, C#, Go, PHP/Laravel; PostgreSQL, MySQL, Redis, Cassandra; REST, Kafka

Frontend: React, Next.js, Angular, JavaScript, TypeScript

Cloud & delivery: Azure, AWS, Docker, Kubernetes, Terraform, CI/CD

Applied AI: LLM APIs, LangGraph, RAG, embeddings, vector databases

PROFESSIONAL EXPERIENCE

Duplo — Senior Full Stack Engineer

Jul 2025 – Present

Nigeria · Hybrid · Full-time

- Develop the internal Admin Portal for operations teams managing businesses, transactions, fees and compliance workflows.
- Implement authentication, role-based access control and maker-checker approvals for secure, auditable administrative actions.
- Build expense-management and payment-platform integrations with upstream and downstream services, applying defensive programming and resilient handling of partial availability.

Bitso — Senior Software Engineer

Apr 2023 – Jul 2025

Mexico · Remote · Contract

- Developed and integrated an AI customer-support assistant using LLM APIs and LangGraph to answer natural-language queries.
- Designed agent workflows for transaction risk analysis, using semantic embeddings, vector retrieval and RAG to surface transaction information and flag suspicious patterns.
- Shipped Java and PHP/Laravel services for high-growth markets serving hundreds of thousands of users.

Microsoft — Senior Software Engineer

Dec 2021 – Feb 2023

Nigeria · Hybrid · Full-time

- Reduced API scan completion time from 15 hours to 5 hours through automated processing.
- Automated runtime and documentation issue logging and background workflows using Azure DevOps, Service Bus, Durable Orchestration Functions, Worker Service and Application Insights.
- Built distributed web applications using C#, Python, TypeScript and Node.js, and implemented Azure CI/CD pipelines.

Indeed — Senior Software Engineer

Jun 2021 – Sep 2022

United States · Remote · Contract

- Maintained and upgraded an artifact-distribution system using Java, Go, Cassandra, MySQL, MongoDB, Consul and ZooKeeper.
- Migrated services from on-premise infrastructure to AWS using Python and upgraded services to run on Kubernetes, with Docker-based delivery.

Interswitch Group — Software Engineer

May 2018 – Nov 2021

Nigeria · Hybrid · Full-time

- Developed frontend and backend systems for urban mass transit: automated fare collection, ticketing and real-time fleet tracking.
- Delivered a web demonstration within six months, contributing to the company's first Lagos Metropolitan Area Transport Authority (LAMATA) contract.
- Refactored Enterprise Service Bus infrastructure into microservices, with reported uptime increasing from 88% to 99.7%.
- Developed an API gateway providing analytics, security and API management.

Zone — Software Engineer

Jan 2017 – Mar 2018

Nigeria · On-site · Full-time

- Architected and delivered software for the Deja-Vu Platform; performed code reviews, unit and integration testing, and produced system specifications and technical documentation.

Earlier: Embedded Systems Engineering Intern, ElectroHyve Research Institute of Technology · Jun–Nov 2015. Hardware development, component testing and system analysis.

SELECTED INDEPENDENT PROJECTS

LedgerDesk — Payment operations with independent approvals, PostgreSQL leased workers, provider reconciliation and durable attempt history.

RelayDesk — Artifact-delivery simulation and separate local HTTP worker lab with SHA-256 verification, durable SQLite jobs and recovery tests.

SourceDesk — Evidence-based support with pgvector retrieval, source snapshots and handoffs; 40 authored regression cases and five live smoke cases.

TrafficDesk — Deterministic transit simulation with disruption controls, passenger journeys and conservation tests.

EDUCATION & ACHIEVEMENTS

Federal University of Technology, Akure

BE, Electrical & Electronics Engineering · First Class Honours · CGPA 4.53/5

Gold Medal, American Mathematics Competition (2011) · Bronze Medal, International Junior Science Olympiad (2010) · Represented Nigeria, International Mathematical Olympiad (2011).